

Avinash R

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Profile

EEE student at SSN College of Engineering with a strong passion for robotics, embedded systems, and product development. Driven to build real-world engineering solutions that tightly integrate hardware and software — from PCB design to autonomous navigation algorithms.

Education

SSN College of Engineering B.E. Electrical and Electronics Engineering	2024 – Present
PSBB Millennium School, Gerugambakkam Class XII · 92.2%	2024

Achievements

- **1st Prize** — Line Follower Robot, *IIT Madras RoboRun*
- **1st Prize** — Line Follower Robot, *CEG Kurukshetra*
- **2nd Prize** — Maze Navigator, *REC Tittanium*
- Cleared **Stage 2** — E-Yantra Robotics Competition, *IIT Bombay*

Projects

Foldable Electric Scooter — Hardware · BLDC Control
Designed and developed a compact electric scooter integrating BLDC motor control and multi-mode acceleration, with a strong focus on real-world usability and structural compactness.
Maze Solver Robot — Robotics · Algorithm
Built an autonomous robot implementing the Flood Fill algorithm for efficient grid navigation and shortest-path optimisation.
Line Follower Robot — ESP32 · PID Control
Engineered a high-precision line-tracking robot using an ESP32 microcontroller with a tuned PID control loop for stable, accurate path following.
Smart Home Pass System — Embedded · Web
Developed a secure digital outpass system for hostels, featuring a streamlined approval workflow with role-based access.
Voice-Controlled Robot — Speech · Embedded
Designed a robotic system with real-time directional control via voice commands, integrating speech recognition with embedded motor drivers.
Home Automation System — IoT · Embedded C
Implemented an embedded IoT system enabling remote monitoring and control of household appliances via a lightweight communication interface.

Skills

Programming	C++, Python, Embedded C, Assembly
Microcontrollers	Arduino UNO, ESP32, STM32, 8051, 8085
Tools	Fusion 360, MATLAB, CoppeliaSim, STM Cube IDE, Arduino IDE, VS Code, KiCad, AutoCAD, LTSpice, Keil µVision
Core Skills	PCB Design, 3D Design, Robotics, Soldering, Welding
Algorithms	PID, Flood Fill, LSRB